

Implementing Verdier's robust extrapolation in the CKT model

Relaxing the i.i.d. migration-cost assumption

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Our current estimator is Verdier's “simple” extrapolation

We currently follow Suri (2011) and Tjernström (2023):

- impose the LCA restriction $\Delta_i = \beta + \phi \theta_i$
- identify ϕ from switcher trajectories
- extrapolate to non-migrants (d_N, d_T) along the LCA line

Verdier calls our approach the *simple extrapolation* & proposes a *robust* alternative.

The simple extrapolation assumes i.i.d. migration-cost shocks

Our current Assumption A2 states that the non-pecuniary utility shock

ν_{it}^l is i.i.d. across individuals i , time t , and locations l .

The migration-cost shifter is independent of baseline skill and comparative advantage **everywhere in the sample**.

This rules out any systematic difference in migration-cost distributions across regions, institutions, or subpopulations.

Practical reading

Two workers in different parts of the country face the same cost distribution, conditional only on their own comparative advantage.

The migration literature documents exactly what A2 rules out

Regional variation in migration cost is the central fact of this literature:

- **Bryan & Morten (2019):** Indonesian migration costs near the world misallocation frontier.
- **Lagakos et al. (2020):** migration costs vary substantially within and across developing countries.
- **Pulido & Swiecki (2021):** sector frictions correlate with regional productivity.
- **China specifically:** hukou generates large, regionally-varying barriers.

Our current model rules this out by assumption.

Verdier's robust extrapolation weakens A2 to a conditional version

Decompose the migration-cost shock:

$$\nu_{it}^l = m_l(v_i) + \tilde{\nu}_{it}^l,$$

where v_i is an observed, time-invariant individual-level index.

Only the residual $\tilde{\nu}_{it}^l$ is i.i.d. *conditional on* v_i . The region-level component $m_l(v_i)$ can correlate arbitrarily with comparative advantage.

We use first-wave province (all three countries).

Why this is weaker

A2 required *unconditional* independence. A2' requires only *within-province* independence.

The LCA gets a province-specific intercept

Slope stays common; intercept varies.

Simple:

$$\Delta_i = \beta + \phi \theta_i.$$

Robust:

$$\Delta_i = \beta(v_i) + \phi \theta_i.$$

- Slope ϕ remains common—this is still the LCA restriction.
- Intercept $\beta(v_i)$ absorbs province-level cost heterogeneity.
- ϕ identifies off within-province switcher variation.
- Extrapolation to non-migrants happens province-by-province, then averaged.

Pooling provinces mixes differentially-selected switcher samples

Consider two provinces:

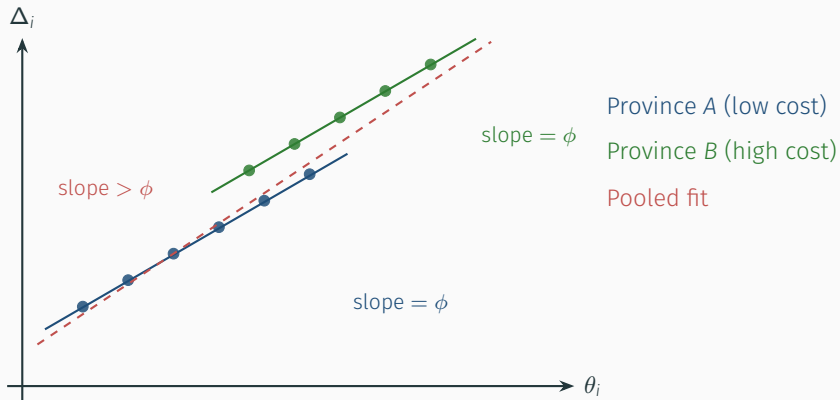
- **Low-cost province** (coastal). Many types migrate; switcher sample covers a wide range of θ_i .
- **High-cost province** (remote). Only high-return types migrate; switcher sample is skewed.

Pooled regression: biased by across-province difference in selection.

Within-province: clean slope, holds migration cost constant.

Partialling out v_i purges this selection bias.

Within-province slopes agree—the pooled slope is pulled off



Each province's switcher cloud has true slope ϕ . The pooled line tilts because of intercept differences, not because ϕ varies.

Implementation adds v_i -interacted means to the GRC regression

Currently:

$$y_{it} = \sum_{\underline{d}} \mu_{\underline{d}} \mathbb{1}\{\underline{d}_i = \underline{d}\} + \Delta_{\underline{d}_0} D_{it} + \sum_{\underline{d} \in \mathcal{D}_S \setminus \underline{d}_0} \phi(\mu_{\underline{d}} - \mu_{\underline{d}_0}) D_{it} \mathbb{1}\{\underline{d}_i = \underline{d}\} + \cdots + \varepsilon_{it}.$$

Under robust: trajectory means $\rightarrow$ trajectory $\times$ province means.

$$y_{it} = \sum_{\underline{d}, v} \mu_{\underline{d}, v} \mathbb{1}\{\underline{d}_i = \underline{d}, v_i = v\} + \Delta_{\underline{d}_0} D_{it} + \phi \sum_{\underline{d} \neq \underline{d}_0} (\mu_{\underline{d}, v_i} - \mu_{\underline{d}_0, v_i}) D_{it} \mathbb{1}\{\cdot\} + \cdots$$

- Slope ϕ unchanged—still a single scalar.
- Standard errors cluster at v_i .
- Stayer return Δ_{d_N} averaged over clusters with switcher support.

The cost-shifter diagnostic is a falsification test, not a substitute

Verdier proposes a diagnostic via observed continuous cost shifter w_i :

$$\theta_i^* = \alpha_0 + \alpha_1 \Delta_i + \alpha_2 w_i + e_i, \quad H_0 : \alpha_2 = 0.$$

In Kenya he uses w_i = average distance to nearest seed seller.

- Requires an observed exogenous cost shifter.
- Falsifies the simple extrapolation if $\alpha_2 \neq 0$.
- Does not deliver an alternative estimate.
- Not required for the robust extrapolation, which uses v_i .

We focus on v_i = province; cost-shifter test is future robustness.

The move buys a weaker assumption at tractable cost

We gain:

- $A2'$ matches what the migration literature documents.
- The CHN hukou split becomes a principled special case ($v_i = \text{hukou}$) rather than an ad hoc fix.
- Stayer ATE estimates defensible against Verdier's concern in Suri's data.

We give up:

- Δ_{d_N} averages over provinces with switcher support, not the full population.
- Thinner identification of ϕ : only within-province switcher variation contributes.
- Cluster-robust standard errors at the v_i level.

Feasibility survey first, then implementation

- Survey **data/processed/** for each country: available baseline geographic variables, unique values per granularity, switchers per cluster, always-rural support fraction.
- Implement v_i = first-wave province across all three countries in **run_grc**.
- Report simple and robust estimates side by side; run Verdier's overidentification test of the LCA line.
- Check whether raw CFPS/IFLS/TZNPS files record province of origin or birth; if so, use as a robustness specification.